

Scientific Computing 2026

Week 5: Sep 28, 30

Nonlinear Optimization

September 16, 2026

This week we study optimization of continuous nonlinear problems. Monday we cover *unconstrained* optimization, i.e., when we minimize a function $f: \mathbb{R}^m \rightarrow \mathbb{R}$ and allow minima to be found anywhere in $\mathbb{R}^m$. Wednesday we cover two topics: First, *constrained* optimization, in which minima are sought in some restricted subset $S \subset \mathbb{R}^m$ defined by a set of equations $\mathbf{g}(\mathbf{x}) = \mathbf{0}$, $g: \mathbb{R}^m \rightarrow \mathbb{R}^n$. Second, we look at *metaheuristics*: schemes that can often help us to find better solutions. (An *algorithm* is a step-by-step procedure that is guaranteed to produce the correct result after a finite number of steps, given its preconditions are satisfied. A *heuristic* is like an algorithm, except that you are not guaranteed a result. A *metaheuristic* is a template for building heuristics for many different types of problems.)

Monday: Methods from the Book

Topics: Intro to nonlinear optimization
Optimization in 1D
Unconstrained optimization in n D:
Steepest descent
Newton and Quasi-Newton (BFGS)
Conjugate gradients

Read beforehand: Chapter 6: 6.1–6.2.2, 6.4–6.5.6

Wednesday: Constrained Optimization and Metaheuristics

Topics: Constrained optimization
Lagrange multipliers
Metaheuristics

Read beforehand: Chapter 6: 6.2.3, 6.7

Read afterwards: Metaheuristics papers: links will be on Absalon